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क्षणिक Kshanik

"kṣaṇik" — transient, momentary · Physics-Informed Yield Surrogate for a Non-Isothermal Plug-Flow Reactor

A machine learning surrogate for reactor yield prediction, built by fitting the governing reaction kinetics and energy balance directly, then applying **Gaussian Process** and **CatBoost** models to correct only the residual error the mechanistic fit leaves behind.

Cross-validated RMSE 8.79 on 150 training observations

Team Transience · Fugacity 2026 ML Hackathon, IIT Kharagpur

· Vaishnav Aditya · Preenithi Varshana · Kritika Pandey · 150 training rows, 50 held-out test rows

`train_dataset.csv` → `mechanistic_pfr.py` → warped GPR / CatBoost → `overall_yield`

[GitHub Repository ↗](#)[Full Technical Report \(PDF\) ↗](#)

25%

ROWS AT EXACT ZERO
YIELD

5 → 13

RAW INPUTS → PHYSICS
FEATURES

8.79

BEST CV RMSE (WARPED
GPR)

2

MODELS: SMALL-DATA
& LARGE-DATA

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LIMITATION OF DIRECT REGRESSION

Black-Box Models Cannot Isolate the Yield Collapse

A regressor fit directly on `flow_rate`, `concentration`, `inlet_T`, `length`, and `jacket_T` must infer, from 150 rows alone, that yield falls to exactly **0%** whenever residence time and temperature drive the side reaction ($B \rightarrow C$) past the desired one ($A \rightarrow B$). No single raw feature correlates strongly with yield — the highest is $|r| = 0.64$ — because the governing variable is a non-linear combination the model has no basis to construct from data this limited.

TRANSCIENCE APPROACH

A Mechanistic Prior, Corrected by a Gaussian Process

We fit a non-isothermal PFR model directly — series kinetics $A \rightarrow B \rightarrow C$ under Arrhenius rate laws, coupled to a jacket energy balance — using nonlinear least squares on the training set. Its output serves as the **prior mean** for a Gaussian Process, which is left to learn only the mechanistic model's residual error rather than the full yield surface.

MODEL PERFORMANCE — 5×10 REPEATED K-FOLD CV

MECHANISTIC
MODEL ALONE**12.17**RMSE · physics
only, no MLADDITIVE-
RESIDUAL GPR**10.14**RMSE · mechanistic
mean + plain GPWARPED-LOGIT
GPR**8.79**RMSE · selected
small-data modelCATBOOST +
SYNTHETIC**11.35**RMSE · large-data
model

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1

FIT ODE

Arrhenius kinetics and energy balance, fit by least squares on the training set

2

ENGINEER FEATURES

Damköhler numbers, selectivity ratio, residence time, ΔT derived from the fit

3

WARP TARGET

Logit transform of yield/100, respecting the $[0, 100]$ boundary

4

LEARN RESIDUAL

Matérn-kernel Gaussian Process fits the mechanistic model's error

5

PREDICT

Inverse transform back to overall_yield for the test set

THE GOVERNING SYSTEM — FIT DIRECTLY TO TRAINING DATA

MOLE BALANCES

$$\begin{aligned} dC_A/d\tau &= -k_1(T) C_A \\ dC_B/d\tau &= k_1(T) C_A - k_2(T) C_B \end{aligned}$$

ARRHENIUS KINETICS

$$k_i(T) = k_{0,i} \exp(-E_{a,i} / RT)$$

$i \in \{1, 2\}$ — one per reaction

ENERGY BALANCE

$$\begin{aligned} dT/d\tau &= [U_a(T_j - T) \\ &\quad + (-\Delta H_1)r_1 \\ &\quad + (-\Delta H_2)r_2] / \rho C_p \end{aligned}$$

jacket heat exchange + reaction enthalpy of both steps

Eight effective parameters — $k_{0,1}$, $E_{a,1}$, $k_{0,2}$, $E_{a,2}$, U_a , ΔH_1 , ΔH_2 , ρC_p — are recovered from the 150 training rows via multi-start nonlinear least squares, not assumed from literature values. The fitted result: $E_{a,2} > E_{a,1}$, meaning the side reaction accelerates with temperature faster than the desired one — the direct physical cause of the zero-yield cliff seen in the training data.



MECHANISTIC FOUNDATION

Non-Isothermal PFR Model

Series reaction $A \rightarrow B \rightarrow C$ under Arrhenius kinetics $k_1(T)$, $k_2(T)$, coupled to a jacket energy balance with explicit reaction-heat terms (Fogler, *Elements of Chemical Reaction Engineering*, Ch. 8). Fit as a grey-box model via multi-start nonlinear least squares, not assumed from literature constants.

Train RMSE: **12.17**

- $E_{a2} > E_{a1}$, consistent with side-reaction dominance at high temperature

SMALL-DATA MODEL

Warped Gaussian Process

The GP is trained in $\text{logit}(\text{yield}/100)$ space rather than on the raw residual. This is the statistically appropriate likelihood for a target bounded on $[0, 100]$ with substantial mass at the boundary, consistent with the censored-regression and warped-GP literature.

CV RMSE: **8.79** $\pm$ 3.87

- outperforms both additive-residual and classifier-gated variants

LARGE-DATA MODEL

CatBoost with Synthetic Augmentation

5,000 synthetic observations, sampled across the operating envelope and simulated through the fitted ODE, are combined with the real data at a $0.15\times$ training weight. A classifier stage first separates the collapsed-yield regime before a regressor is applied, a strategy intended to scale to a larger production dataset.

CV RMSE: **11.35** • versus 15.24 on real data alone, no augmentation